

which is rewritten as

$$\begin{aligned} m_A \|u - u^h\|_V^2 &\leq \langle Au - Au^h, u - v^h \rangle + \langle Au, v^h - u \rangle + \langle Au, v - u^h \rangle \\ &\quad + \langle Au, u - v \rangle + \langle Au^h, u^h - v^h \rangle. \end{aligned} \quad (4.40)$$

Applying (4.2),

$$\langle Au, u - v \rangle \leq \Phi(v) - \Phi(u) + I_\Delta(\psi^0(\gamma_\psi u; \gamma_\psi v - \gamma_\psi u)) - \langle f, v - u \rangle.$$

Applying (4.16),

$$\langle Au^h, u^h - v^h \rangle \leq \Phi(v^h) - \Phi(u^h) + I_\Delta(\psi^0(\gamma_\psi u^h; \gamma_\psi v^h - \gamma_\psi u^h)) - \langle f, v^h - u^h \rangle.$$

Using these inequalities in (4.40), after some rearrangement of the terms, we have

$$m_A \|u - u^h\|_V^2 \leq \langle Au - Au^h, u - v^h \rangle + R_u(v^h, u) + R_u(v, u^h) + I_\psi(v, v^h), \quad (4.41)$$

where

$$R_u(v, w) := \langle Au, v - w \rangle + \Phi(v) - \Phi(w) + I_\Delta(\psi^0(\gamma_\psi u; \gamma_\psi v - \gamma_\psi w)) - \langle f, v - w \rangle, \quad (4.42)$$

$$\begin{aligned} I_\psi(v, v^h) &:= I_\Delta(\psi^0(\gamma_\psi u; \gamma_\psi v - \gamma_\psi u) + \psi^0(\gamma_\psi u^h; \gamma_\psi v^h - \gamma_\psi u^h)) \\ &\quad - I_\Delta(\psi^0(\gamma_\psi u; \gamma_\psi v^h - \gamma_\psi u) + \psi^0(\gamma_\psi u; \gamma_\psi v - \gamma_\psi u^h)). \end{aligned} \quad (4.43)$$

Let us bound the first and the last two terms on the right hand side of (4.41). First,

$$\langle Au - Au^h, u - v^h \rangle \leq L_A \|u - u^h\|_V \|u - v^h\|_V.$$

By the modified Cauchy-Schwarz inequality (2.12), for any $\epsilon > 0$ arbitrarily small,

$$\langle Au - Au^h, u - v^h \rangle \leq \epsilon \|u - u^h\|_V^2 + c \|u - v^h\|_V^2 \quad (4.44)$$

for some constant c depending on ϵ . Applying the subadditivity of the generalized directional derivative,

$$\psi^0(z; z_1 + z_2) \leq \psi^0(z; z_1) + \psi^0(z; z_2) \quad \forall z, z_1, z_2 \in \mathbb{R}^m,$$

we have

$$\begin{aligned} \psi^0(\gamma_\psi u; \gamma_\psi v - \gamma_\psi u) &\leq \psi^0(\gamma_\psi u; \gamma_\psi v - \gamma_\psi u^h) + \psi^0(\gamma_\psi u; \gamma_\psi u^h - \gamma_\psi u), \\ \psi^0(\gamma_\psi u^h; \gamma_\psi v^h - \gamma_\psi u^h) &\leq \psi^0(\gamma_\psi u^h; \gamma_\psi v^h - \gamma_\psi u) + \psi^0(\gamma_\psi u^h; \gamma_\psi u - \gamma_\psi u^h). \end{aligned}$$

Thus,

$$\begin{aligned} I_\psi(v, v^h) &\leq I_\Delta(\psi^0(\gamma_\psi u^h; \gamma_\psi v^h - \gamma_\psi u) - \psi^0(\gamma_\psi u; \gamma_\psi v^h - \gamma_\psi u)) \\ &\quad + I_\Delta(\psi^0(\gamma_\psi u; \gamma_\psi u^h - \gamma_\psi u) + \psi^0(\gamma_\psi u^h; \gamma_\psi u - \gamma_\psi u^h)). \end{aligned}$$

By (4.8) and (4.12),

$$\begin{aligned} I_\Delta(\psi^0(\gamma_\psi u; \gamma_\psi u^h - \gamma_\psi u) + \psi^0(\gamma_\psi u^h; \gamma_\psi u - \gamma_\psi u^h)) &\leq \alpha_\psi I_\Delta(|\gamma_\psi u - \gamma_\psi u^h|^2) \\ &\leq \alpha_\psi c_\Delta^2 \|u - u^h\|_V^2. \end{aligned}$$